

Khawaja Yunus Ali University

School of Science and Engineering

B.Sc. in CSE & EEE

Lesson Plan

Course Title : Physics - I

Course Code : CSE 0533 1101 (CSE) & PHY 0533 1101 (EEE)

Term: Spring – 2024	Date: 01.01.2024	Time: 10.00 AM	Room: 327	Activity: Lecture 1
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Topics to be discussed: The Electricity

Concept of electric charge, conductors and insulators

Target CLO & PLO : CLO1,2,4 & PLO1,2,4

Learning Outcomes : At the end of the class the students will be able to:

- ❖ Define electric charge and its origin, unit of charge
- ❖ Explain how to quantize electric charge
- ❖ Explain some natural phenomenon like lightning or thundering

Methods of Teaching:

Mode/Activities	Time
Introductory Speech	15 min
Delivery of the lesson plan & OBE course outline	5 min
Interactive discussion	10 min
Lecture Presentation on Topics	30 min
Group activity	15 min
Feedback session	10 min
Others = 5 min	5min

Assessment Strategy : Quizzes/Class tests, Class Participation

Sample Questions (MCQ):

- 1) What is the minimum charge of our universe?
a) $1.6 \times 10^{-19} \text{C}$ b) $3.2 \times 10^{-19} \text{C}$ c) $6.4 \times 10^{-19} \text{C}$ d) $9.1 \times 10^{-19} \text{C}$ e) $12.5 \times 10^{-19} \text{C}$
- 2) Which is possible amount of charge of an object?
a) $Q=100.8e$ b) $Q=80e$ c) $Q=13.5e$ d) $Q=9.2e$ e) $Q=4.5e$

Sample Questions (SAQ):

- 1) Define lightning and thunder
- 2) What is the charge conservation principle?
- 3) What is the unit of charge? Define it.
- 4) How is the charge quantized? – explain it.

Reference Books: 1. Giasuddin ahmed – Physics for Engineers (Part-2)
 2. Huq. Rafiqullah- Electricity and Magnetism

Pre-reading book for next Lecture:

- Md. Ziaul Ahsan – Physics for Engineering (Lecture Series), Pages: 326-331
- Dr. Amir Hussain Khan – Physics 2nd Paper (HSC Text Book), Pages: 37-86